

From Governance Reviews to Task Substitution

DGF-Bench and the Role of Forward Deployed Engineers

Jeremy Canale

contact@jeremycanale.com

<https://www.jeremycanale.com>

September 2026

Abstract

Enterprise governance reviews transform project evidence into decisions, conditions, and actions. We evaluate whether tool-using language models can perform this work under explicit policies, using DGF-Bench: 300 synthetic Buy, Integrate, and Build projects with generated documents, architectures, simulated records, and sequential review gates. Agents have access to authoritative structured facts as well as documents. Across 899 evaluable model-project runs, Gemini 3.8 Flash, GPT-5.6 Luna, and DeepSeek v4.1 Flash achieve strict gate success of 94.98%, 83.29%, and 74.18%, respectively. Complete-route success is 76.92%, 42.33%, and 24.67%; recorded API cost totals USD 87.02. Gemini has 100% disposition accuracy: all its strict failures concern evidence conformity. We therefore distinguish decision quality from lexical traceability instead of interpreting the gate-route gap as a general reasoning failure. These results support the technical prospect of automating the represented review tasks. We formulate a conditional task-substitution hypothesis and specify how forward deployed engineers could connect review automation to operational evidence, authority, exception handling, and measurement of human work. A compact labor account shows why benchmark accuracy is not a staffing-reduction estimate. Independent human comparison, semantic evidence adjudication, scaffold-removal tests, and a prospective enterprise cohort are specified as follow-up studies, not completed experiments. The hypothesis of 80% fewer required DGF FTE by 2033 is retained as a field-testable projection with an explicit historical-baseline requirement. All benchmark sources, traces, and datasets are public. This manuscript reorganizes an existing experiment; it reports no new model trials.

Keywords: governance reviews; AI agents; forward deployed engineering; benchmark; task substitution; human workload.

1 Introduction: a gate is a contract, not a profession

A proposed enterprise project is examined by several groups before it proceeds. A legal reviewer checks contractual prerequisites; an architecture reviewer checks the technical design; security and readiness reviewers check protections and operational evidence. Each review consumes information and produces a decision with reasons, conditions, and obligations. The institutional label of the reviewer does not determine whether this transformation must always be performed by a person. The useful unit of analysis is the review contract: the inputs, rule, output, evidence, and authority that make a particular review acceptable.

This observation motivates a task-substitution hypothesis. If software can satisfy a review contract and its operational assurance requirements, it can replace some of the human execution of that review. If it does so across a connected process while reducing total human work, the same volume of governed projects may require fewer staff. Neither implication requires abolishing the gate or lowering its standard. Nor does successful execution of a synthetic contract establish that every enterprise contract has already been captured.

The empirical question in this paper is specific: how reliably do three model endpoints apply explicit governance rules to synthetic dossiers, investigate with tools, produce evidence-bearing decisions, and continue through a sequence of gates? The experiment provides a useful early measurement because it exposes what an agent must return and retains what it actually read and did. It is also deliberately scaffolded: decision-relevant structured facts and policy rules are available to the model. Success is evidence about this policy-application task, rather than a direct comparison with unaided professional judgment.

1.1 Contributions and claim structure

The paper has three contributions. First, it documents a released benchmark with 300 distinct projects, 899 evaluable model–project runs, component-level scores, paired comparisons, and recorded costs. Second, it separates decision agreement, strict evidence conformity, and whole-route success, including concrete examples where their interpretations differ. Third, it defines an implementation and measurement agenda linking task automation to human work: the engineering artifacts an FDE must build, a minimal labor account, and follow-up studies that could substantiate or reject the workforce hypothesis.

Table 1: Three different claim levels. A stronger claim needs an additional measurement.

Level	Claim in this paper	Evidence required
Measured performance	Models often satisfy specified synthetic review contracts.	Released submissions, traces, scores, and uncertainty conditional on the dataset.
Deployment hypothesis	Comparable review tasks can be executed with less total human work.	Independent review quality and measured human hours in operational workflows.
Workforce projection	Required DGF FTE may fall by 80% by 2033 at comparable output.	A defined cohort, credible baseline, all-in labor boundary, and endpoint observations.

The second and third levels are research hypotheses. We do not call eventual complete automation a law: without a deadline or a specified comparison population, that endpoint cannot be rejected by a finite observation period. The useful empirical target is a change in required human work over a stated period with quality preserved. A substantial reduction can occur while some human tasks remain necessary.

1.2 Relation to prior research and to the companion manuscript

WorkArena evaluates agents performing enterprise-software tasks (Drouin et al., 2024). DGF-Bench focuses instead on connected governance reviews with explicit decisions, supporting findings, authority, and evidence. AgentDojo evaluates tool-using agents exposed to untrusted data and prompt injection (Debenedetti et al., 2024); the present run does not provide that adversarial evaluation. These comparisons locate the task rather than establish superiority across benchmarks.

The workload interpretation follows a task-level distinction between automation and aggregate employment: displacement of particular tasks can coexist with new tasks and changing demand (Acemoglu and Restrepo, 2019). Bainbridge’s analysis of automation motivates explicit attention to the work left for human operators (Bainbridge, 1983). Field evidence such as Brynjolfsson et al. (2025) illustrates a different empirical object: productivity observed in an operating organization. We do not transfer a productivity effect from another occupation to DGF reviewers or infer employment effects from model scores.

This is a shorter, empirical companion to *The Last Human Gate: Forward Deployed Engineering and the Automation of Enterprise Governance*, by the same author. Both manuscripts use the same September 2026 DGF-Bench experiment; they are not independent replications. This paper

removes the universal-law framing, reduces the scenario apparatus, and makes the validation gap an operational research plan. The original manuscript and data remain available separately.

The routes are author-designed synthetic workflows informed by practitioner experience. Their definitions, generated artifacts, rule implementation, and scoring interface are public, so the benchmark specification can be inspected without access to an unpublished practitioner document. This makes the implemented task inspectable; it does not establish that its rules represent a particular enterprise or an independently validated professional standard.

2 DGF-Bench: measured review performance on 300 synthetic projects

2.1 Question, population, and information available to the agents

This experiment asks whether an AI agent can apply a declared governance policy, identify the required findings and actions, cite observed evidence, respect its authority, and carry a review through a connected route. It measures review decisions and proposed actions, not implementation of remediation or changes in human workload. The field-validation studies proposed later in this paper have not been conducted.

Three practical questions organize the analysis: (1) does the model reach the disposition required by the rules, (2) does it produce all the supporting work needed for a valid review, and (3) does that reliability survive a sequence of reviews at an acceptable API cost? The unit called a *case* or *dossier* is one generated project; a *gate* is one checkpoint on its route. Three models reviewing the same 300 projects produces 900 planned model–case runs, not 900 independent business projects.

The dataset contains 300 fictional projects: 100 Buy, 100 Integrate, and 100 Build cases. Buy and Integrate contain six gate occurrences each, and Build contains five, for 1,700 occurrences per model and 5,100 planned observations across three models. Cases were generated before these evaluations at difficulty setting 4, with sampling conditioned to improve decision coverage. Project context, owners, budgets, data classifications, technical architectures, documents, and simulated system records derive from canonical case facts. Documents can be missing, stale, or inconsistent. Specialist reviews precede a final General review; the routes are examples rather than a universal enterprise standard.

The agent receives gate objectives, a structured output contract, an explicit decision policy, and tools to read allowed evidence and query the synthetic environment. In particular, the policy includes executable rule definitions and identifies an authoritative structured `REVIEW_FACTS` snapshot. The evaluator-only reference file is inaccessible through the agent’s tools. This is nevertheless a strongly scaffolded test of policy application: the agent need not discover the rules or reconstruct every decision-relevant fact from unstructured documents. Architecture images are available under the vision setting; image availability is not evidence that image interpretation was necessary for success. These design choices limit claims about expert reasoning on independently collected enterprise dossiers.

The dataset audit reports 300 distinct project identifiers, architecture signatures, and complete decision-fact signatures, but only 267 distinct whole-case finding patterns. Gate-specific relevant fact signatures include 142 of 200 Compliance cases and 241 of 300 Security cases. All rules applicable to the selected phases are covered according to the generator’s audit. Phase-adjusted normalized decision entropy ranges from 0.9971 to 1.0000. Balanced labels improve coverage, but do not estimate enterprise prevalence or establish independent business validity.

2.2 What is generated, and what the model actually does

The case generator is a program using seeded facts, rules, and document templates. The AI model under evaluation is the reviewer, not the author of the fictional need or architecture. A Buy scenario represents acquiring a solution; Integrate represents connecting existing systems; Build represents developing a new project. Those scenarios are given budgets, users, owners, dates, data classifications, and technical constraints. A generated name is a scenario label, not an independently elicited customer requirement.

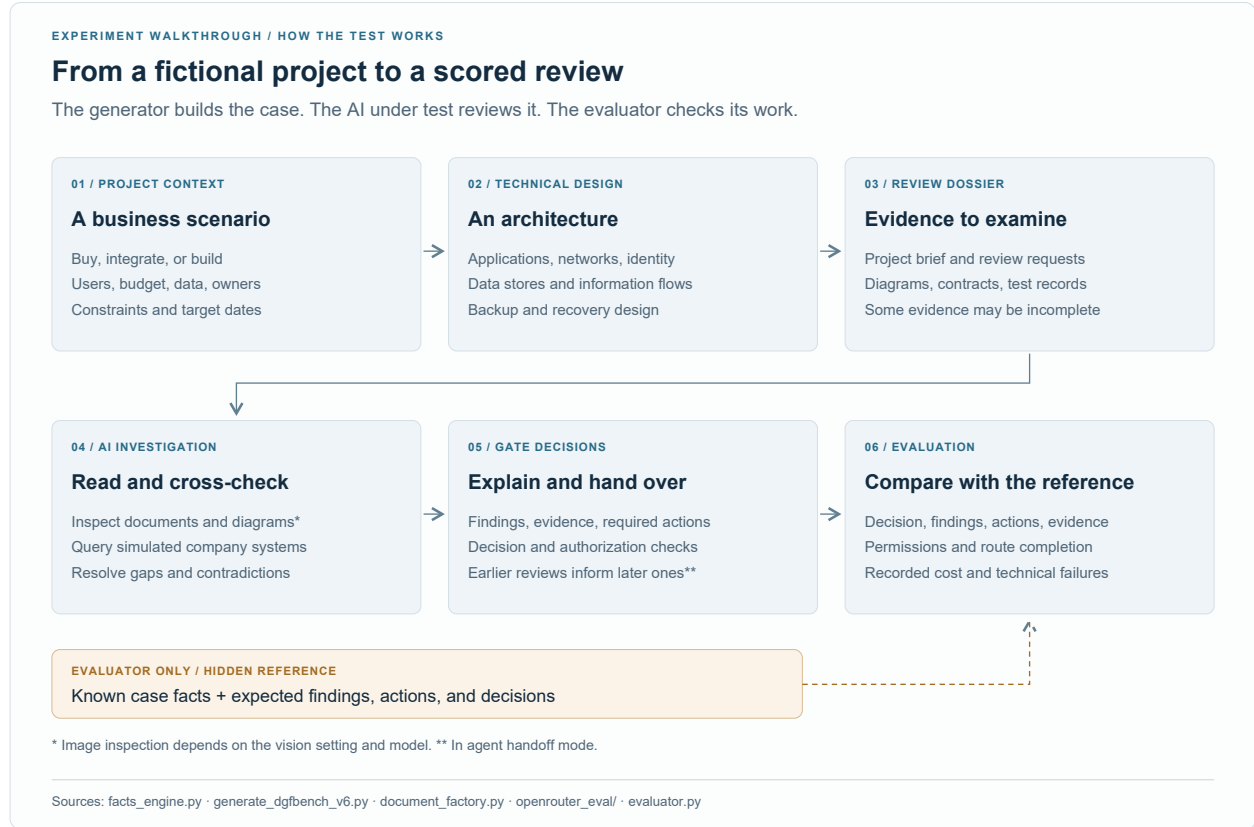


Figure 1: Experimental workflow. The generator constructs the project and evidence; the tested model investigates and submits reviews; the evaluator uses a separate reference. The diagram describes the implemented test, not a measured enterprise deployment.

1. **Construct the project facts.** The program fixes what is true in the synthetic case, such as the requested budget, whether a supplier has supplied a document, or whether a restore test succeeded. Decision-coverage sampling produces cases with different required outcomes.
2. **Generate the design and dossier.** The same case drives an architecture diagram, project charter, gate review requests, and evidence files. A reviewer may receive a technical design, network/data-flow tables, contract records, security evidence, or readiness tests, depending on the gate (Table 2). Some presented evidence is incomplete or stale.
3. **Start a gate review.** The harness supplies the project context, gate and phase, applicable policy, permitted evidence and tools, and the answer schema. The model decides which allowed records to inspect. It can list and read evidence, query simulated systems, request information, and inspect existing authorization mandates.
4. **Submit a decision.** The model returns a disposition, finding identifiers, proposed actions, evidence references and supporting excerpts, an authorization indicator, and a rationale. A typed submission is recorded. Producing a syntactically valid record alone does not satisfy the substantive score.

5. **Continue the route.** In the tested **agent** mode, the harness passes actual upstream model reviews to later checkpoints. The final General review consolidates specialist opinions and its own checks. The tested routes are listed in Table 3.
6. **Evaluate and retain the trace.** A deterministic evaluator compares the submitted record and observed tool events with the reference. It records component scores, full gate and route success, critical misses, false approvals, provider failures, and usage. It is not another LLM judge. Failed or incorrect decisions are retained; infrastructure exclusions are reported.

Table 2: Concrete examples of generated review material. The contents vary with the route and gate. Authoritative structured snapshots are available in addition to these artifacts.

Material	Examples in the case	Review question
Project context	Charter and review request (Word), project context (JSON)	What is proposed, for whom, at what cost, and under whose ownership?
Architecture	Diagram (SVG/PNG), technical design, network and flow tables	Do the design facts meet the declared constraints?
Security	Identity, vulnerability, monitoring, and resource records	Are the protections required by the policy present?
Supplier and legal	Vendor evidence, contract versions, data-processing agreement status	Are the supplier and contractual prerequisites satisfied?
Readiness	Backup, restore/failover and load-test records; runbook	Can the proposal demonstrate the required operational readiness?

Table 3: The three routes actually evaluated; each project ends with General. These are the complete route sequences evaluated in this study.

Route	Cases	Ordered gates
Buy	100	Procurement, Legal, Compliance, Security, IT, General
Integrate	100	IT, Architecture, Security, Legal, Compliance, General
Build	100	IT, Architecture, Security, Tech Readiness, General

The five dispositions are **GO** (proceed), **GO_WITH_RESERVATIONS** (proceed under conditions), **REWORK** (revise the proposal), **SUSPENSION** (wait for a prerequisite), and **NO_GO** (do not proceed as submitted). Approval is therefore not the target outcome for every dossier: a justified refusal is a successful review. Tool actions take place in the synthetic environment; a request to fix a defect is not evidence that the defect was fixed.

2.3 A generated project and a recorded successful review

Case DGF-BLD-035200_build, *Project Falcon—Secure Vendor Portal*, provides a concrete example. Its generated HR context has 1,000 users, a requested budget of EUR 500,000, confidential data, high business criticality, and a primary region of UAE North. Figure 2 reproduces its actual evidence diagram. It depicts identity, application and data services, monitoring and backup, and a secondary recovery region. Its labels are claims in the dossier to be checked, not an independently validated production blueprint.

In Gemini’s recorded Tech Readiness review, the authoritative record has a draft runbook. The agent identifies TR-OPS-001 and proposes completing the operational handover. Other required readiness checks pass. It cites the observed excerpt "**runbook_status**": "**draft**", consults the active mandate, and executes the simulated conditional-approval tool with a condition to approve the runbook before production cutover. The tool reports **EXECUTED** and explicitly leaves the finding unresolved. The submitted outcome is therefore **GO_WITH_RESERVATIONS**, with the finding and required action retained and the authorization indicator set to true.

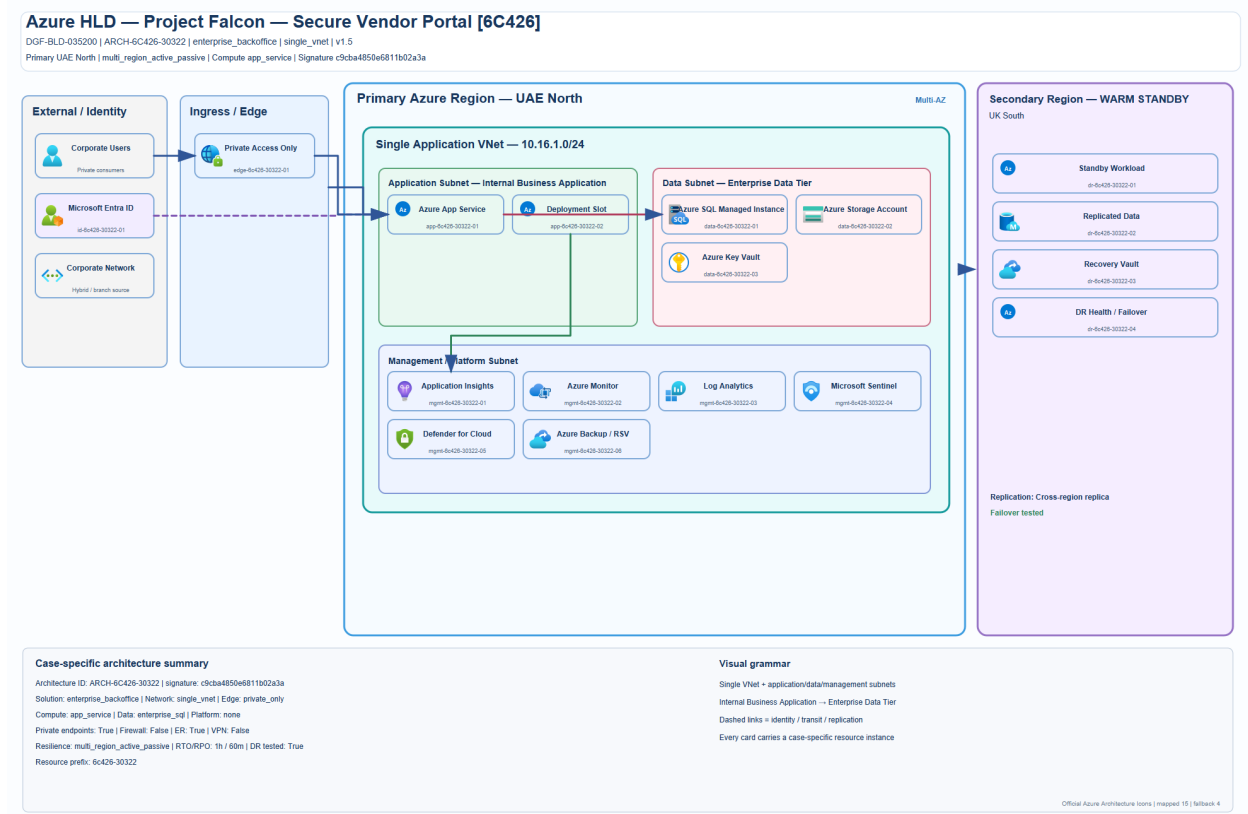


Figure 2: An actual generated architecture document from Project Falcon. Full-resolution evidence is in the released dataset. The figure illustrates what a dossier can contain; it does not imply that this image was necessary to obtain the correct review outcome.

All three models pass all five gates of this particular dossier. Their observed dispositions are GO, GO, GO, GO, GO_WITH_RESERVATIONS, and GO in route order. This example was selected after evaluation to explain the mechanics, not as a representative sample or an additional experiment. The study-level counts include every evaluable case, including unsuccessful reviews. The public research directory includes `worked_examples.json`, with the original archive-relative paths and selected recorded outputs for this example and the evidence-failure example below. The complete traces remain in the release.

2.4 Protocol, completion, and outcomes

Table 4 identifies the three tested endpoints. Each receives the same case population, policy, tool interface, and task limits; no model is trained or fine-tuned by this experiment. The comparison concerns these deployed endpoints under the recorded conditions, not different implementations of a human review process. Configuration was temperature 0, vision auto, agent-produced handoffs, at most 20 turns and 40 tool calls per gate, and 8,192 output tokens per turn. Gates remained sequential within a project. The last resume used 12 workers, four per model, and a USD 110 cost cap; an earlier invocation used six workers and a USD 65 cap. Data collection spanned 22–23 September 2026. Resumption retained completed cases and valid gate checkpoints. Historical provider/key-limit failures remain in the released traces. The 78.73-minute elapsed time in the last run statistics describes that resume only.

Of 900 planned model-case runs, 899 are evaluable: 300 for DeepSeek, 299 for Gemini, and 300 for Luna. Gemini case DGF-INT-035128_integrate failed at its first IT gate with a provider generation error (`finish_reason=error`); it is excluded as an infrastructure failure. Thus 5,094 of 5,100 planned gates are evaluable, with all expected gates attempted in the evaluable cases. There

are no recorded final agent-protocol failures or model-identifier mismatches. Provider-reported identity is not an independent audit of underlying weights. DeepSeek was served by multiple OpenRouter providers; Gemini and Luna were recorded through Google AI Studio and OpenAI.

Table 4: Models used in the experiment. Names are short labels for the exact requested OpenRouter identifiers, not claims about independently verified model weights.

Model label	Requested identifier	Recorded serving provider
DeepSeek v4.1 Flash	deepseek/deepseek-v4.1-flash	Multiple providers
Gemini 3.8 Flash	google/gemini-3.8-flash	Google AI Studio
GPT-5.6 Luna	openai/gpt-5.6-luna	OpenAI

Scoring uses DGF-decision-v8.2-authorized-review. A gate succeeds strictly only if its disposition, finding set, action set, evidence provenance, and authorization indicator all satisfy the reference. Findings and actions require F1 equal to one. Evidence requires an observed reference and exact excerpts supporting the relevant field/value tokens. This is lexical provenance checking, not independent verification of arbitrary reasoning. Authorized conditional approvals can change the effective reference disposition; the evaluator verifies the mandate and recorded action. General combines effective reference opinions, so an upstream error can also cause a downstream failure. Complete-route success requires every gate to succeed strictly; it does not mean that remediation was executed.

For model m , let s_{mcg} be one when gate g of case c passes every component, and zero otherwise. The two principal rates are

$$\text{GateCSR}_m = \frac{\sum_c \sum_{g \in G_c} s_{mcg}}{\sum_c |G_c|}, \quad \text{RouteSuccess}_m = \frac{\sum_c \prod_{g \in G_c} s_{mcg}}{|C_m|},$$

where C_m contains the evaluable cases and G_c their expected gates. This gate rate is weighted by gate count: a six-gate route contributes six observations and a five-gate route contributes five. A model passing four gates of a five-gate case has 80% gate success for that case but zero complete-route success. Decision-only accuracy checks just the submitted disposition. A critical miss is an omitted reference finding classified as critical; a false approval is GO or GO_WITH_RESERVATIONS when the effective reference requires rework, suspension, or refusal.

Gate confidence intervals use a route-stratified bootstrap of whole cases, preserving dependence between a project’s gates (2,000 draws, seed 81931; a case-count Wilson fallback handles degenerate intervals). Route intervals use Wilson bounds. Pairwise differences use the same 299 cases for all models, resampled together within routes (10,000 draws, NumPy seed 81931). These are descriptive intervals conditional on the synthetic sampling design. No preregistered analysis or independently adjudicated ground truth is claimed, and one trajectory per model and case does not estimate run-to-run inference variability.

2.5 Overall and route-level results

Table 5: Measured synthetic benchmark outcomes. Gate CSR requires complete conformity; route success requires all gates. Cost includes recorded failed attempts.

Model	Cases	Gate CSR [95% CI]	Route success [95% CI]	USD
DeepSeek	300	74.18% [71.88, 76.47]	24.67% [20.13, 29.84]	10.09
Gemini	299	94.98% [93.86, 96.04]	76.92% [71.82, 81.34]	71.88
Luna	300	83.29% [81.29, 85.24]	42.33% [36.87, 47.99]	5.05

Gemini succeeds strictly on 1,609 of 1,694 gates, Luna on 1,416 of 1,700, and DeepSeek on 1,261 of 1,700. Corresponding complete-route counts are 230 of 299, 127 of 300, and 74 of 300 (Table 5). Mean weighted component scores, which are not strict success rates, are 99.68%, 97.67%, and 95.02%, respectively. The distinction matters: strong average components can coexist with many incomplete project reviews.

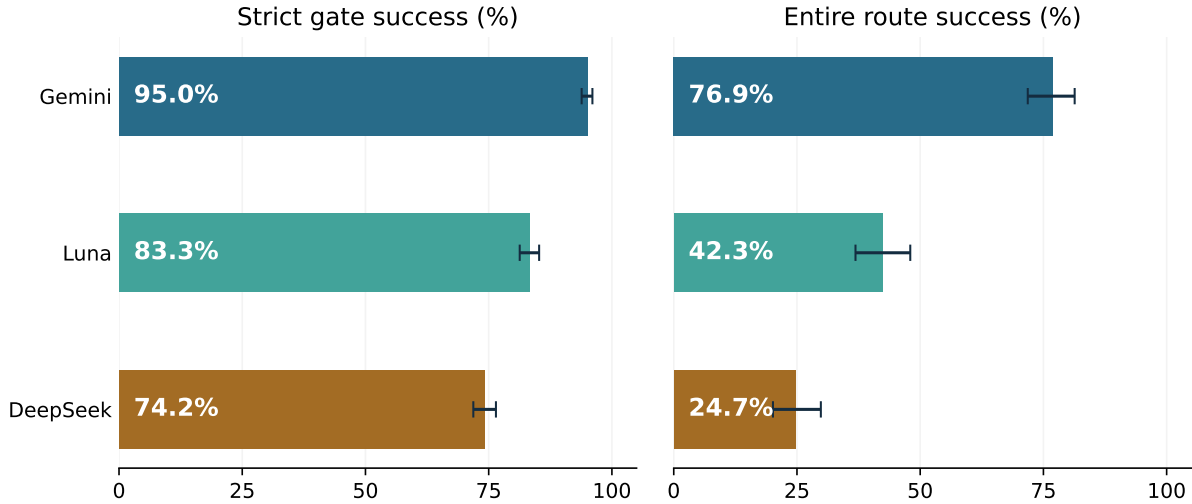


Figure 3: Measured gate and complete-route review outcomes. Intervals are computed over independent case units rather than treating all gates as independent trials. These are model observations on synthetic dossiers, separate from the workforce scenario figures.

On the 299 common cases, CSR is 94.98% for Gemini, 83.23% for Luna, and 74.09% for DeepSeek. Paired differences are 20.90 percentage points for Gemini minus DeepSeek (95% interval 18.30–23.44), 11.75 for Gemini minus Luna (9.62–13.93), and 9.15 for Luna minus DeepSeek (6.67–11.63). The ranking persists on the matched cohort; this is not a universal model ranking.

Table 6: CSR and complete-route counts by project route. Gemini excludes one Integrate case.

Route	DeepSeek: CSR; routes	Gemini: CSR; routes	Luna: CSR; routes
Buy	67.83%; 13/100	97.67%; 87/100	74.67%; 20/100
Integrate	78.17%; 32/100	95.12%; 75/99	87.67%; 52/100
Build	77.00%; 29/100	91.60%; 68/100	88.40%; 55/100

Gemini’s route success falls from 87 of 100 Buy cases to 68 of 100 Build cases. Luna succeeds on 20 Buy, 52 Integrate, and 55 Build cases; DeepSeek succeeds on 13, 32, and 29, respectively (Table 6). These differences do not isolate the causal effect of route length, because gate composition and case content also differ.

Procurement is the lowest-CSR gate for both Luna (33%) and DeepSeek (43%). Luna’s General CSR is 61.67%. Luna exceeds Gemini’s descriptive CSR in Architecture and Tech Readiness (Table 7); these local comparisons are exploratory, without multiplicity-adjusted tests or a claim of specialist superiority.

2.6 Evidence failures, unsafe approvals, and cost

Disposition accuracy alone is 100% for Gemini, 98.12% for Luna, and 95.24% for DeepSeek. All 85 Gemini strict failures concern evidence conformity; findings, actions, dispositions, and authorization indicators match the reference. Evidence is the sole failing component in 221 of 284 Luna failures and 338 of 439 DeepSeek failures. Invalid excerpts and missing or unobserved references are grouped

Table 7: Strict success by gate family (percent). Counts vary by route composition; complete denominators and intervals are supplied in the released tables.

Gate	DeepSeek	Gemini	Luna
Architecture	75.00	85.43	89.50
Compliance	88.00	100.00	96.00
General	68.67	87.96	61.67
IT	87.00	100.00	97.67
Legal	68.50	99.50	80.50
Procurement	43.00	96.00	33.00
Security	72.33	98.66	92.00
Tech Readiness	71.00	89.00	97.00

by these diagnostics; not all are harmless formatting defects. Changing the evidence rule after observing results would change the estimand. A semantic evidence metric would require a separately declared analysis.

A recorded example of a correct decision with a failed proof. In DGF-BLD-035201_build, Gemini reads the Architecture facts: an API gateway is required but absent, and measured latency is 55 ms against a 20 ms target. It correctly names ARCH-API-001 and ARCH-PERF-001, proposes adding the gateway and remediating latency, and submits REWORK. Two attempted conditional approvals are rejected by the tool: one omits an open finding and the other includes a finding ineligible for risk acceptance. The submitted refusal to proceed is consistent with those restrictions.

The gateway excerpt reverses two fields. It places `api_gateway_present` before `api_gateway_required`; the observed record has the reverse order. The values are correct but the excerpt is not an exact observed sequence. Evidence fidelity is 0.5 for the two findings, so strict success is zero while disposition, finding, action, and authorization scores are all one. The weighted component score is 0.95. This selected trace explains exactly how a fluent, substantively correct review can fail this benchmark’s evidence contract; it does not justify silently relaxing that contract after seeing the results.

DeepSeek records one critical finding miss and one false approval, both in the Architecture review of DGF-INT-035136_integrate. An address overlap required NO_GO; at forced finalization, the model submitted GO with an empty finding set, no evidence, and a “placeholder” rationale. The response passed the submission schema but failed the substantive score. This observation combines decision quality with the ability to finalize under resource limits.

Luna records three critical misses and one false approval. In case DGF-INT-035197_integrate, its Legal review approved a dossier with a missing DPA, while its own rationale said that REWORK was required. The other critical misses concern private endpoints in Security and an upstream NO_GO in General. The latter can include error propagation and should not be interpreted as an independent new safety event. Gemini has no observed critical miss or false approval in its evaluable runs; this does not establish zero risk. Recorded truncated-response counts are 2 for DeepSeek, 14 for Gemini, and 0 for Luna, without implying that these counts equal incomplete cases.

The usage ledgers sum to USD 87.015829172, agreeing with the cumulative budget counter. Gemini accounts for USD 71.88, DeepSeek USD 10.09, and Luna USD 5.05, including recorded retries and failures. Approximate cost per evaluated dossier is USD 0.240, 0.034, and 0.017, respectively. Luna therefore combines a higher overall strict score with lower recorded cost than DeepSeek in this execution. Costs depend on provider routing, caching, price, and retry history; they are not a labor-cost comparison or a permanent pricing claim.

3 What the scores establish

3.1 A near-saturated disposition task and a stricter trace contract

Gemini’s 100% disposition accuracy is a strong result under the declared conditions. It also indicates that this sample offers little remaining discrimination on that component for this endpoint. The model can read explicit rules and authoritative structured facts; this experiment does not isolate how much ability is needed to recover those facts from documents. A deterministic policy engine operating on the same permitted facts is therefore an important missing comparator. Without it, the additional contribution of a language model over explicit rule execution is not quantified. Its ability to interact, summarize, and assemble a submission is observed as part of the whole system, not identified by an ablation.

Strict success answers a different question: does the entire submitted review satisfy the implemented contract, including exact evidence provenance? It is useful when downstream processing requires precise traceability, but it is not a direct measure of independent expert judgment. The selected Architecture example shows a correct decision and correct values with a failed excerpt because two fields were reversed. The Legal false approval by Luna and DeepSeek’s placeholder approval illustrate materially different errors. Those classes should not be collapsed into a single narrative about reasoning failure.

Table 8: Separate observed outcomes. Evidence-only failures have no other failing scored component; they are not all established to be harmless formatting errors.

Outcome	DeepSeek	Gemini	Luna
Correct dispositions / evaluable gates	1,619/1,700	1,694/1,694	1,668/1,700
Strictly successful gates	1,261	1,609	1,416
Evidence-only gate failures	338	85	221
Other strict gate failures	101	0	63
False approvals	1	0	1
Critical finding misses	1	0	3

The strict gate-route gap for Gemini is consequently a compounding of evidence-conformity failures across routes, not evidence of incorrect final dispositions. A strict route fails when any component at any gate fails. This is the intended operational definition, but the figure 76.92% must be read with that definition. We retain both the original scoring rule and the component breakdown. Neither retrospectively relaxing the rule nor interpreting every lexical failure as a dangerous decision is justified.

The diagnostics do not provide an independently adjudicated fraction of harmless formatting errors among the 85 Gemini failures. The field-order example establishes that such a failure exists, not that it explains most of them. Missing references, unobserved sources, omitted qualifications, and incorrect substantive support would require separate classification.

3.2 A proposed semantic evidence assessment

A semantic assessment should supplement, rather than replace, the frozen lexical score. Its object would be the relationship between an output claim and evidence the agent actually observed. A correct conclusion based on an unseen file would not pass an observed-evidence requirement. A correctly observed passage that does not support the conclusion would not pass semantic support. Harmless serialization changes should be distinguished from both.

We propose four separately recorded judgments for each required finding: whether the source was available and read before submission; whether the cited span is correctly identified; whether it entails the factual claim with its scope, date, and qualifications intact; and whether the claim is sufficient to support the relevant policy finding. Each receives a supported, unsupported, or ambiguous label

and a written reason. Disputed policy interpretation is recorded separately from a disputed quote. Unsupported observations must not be repaired by showing the adjudicator the hidden reference answer as if the agent had seen it.

Two domain-qualified adjudicators would label the same items independently, with model names and original pass/fail outcomes masked where feasible. A third adjudicator would resolve disagreements under a written rubric. Report raw agreement, a chance-corrected agreement statistic, the disagreement categories, and results both before and after adjudication. Model-generated judging can assist triage only after comparison with the human labels; it must not define its own validity.

For an initial audit, include all strict evidence failures and a probability sample of passing items stratified by model, route, and gate. Use the sampling probabilities for any population estimate. If the goal is a new complete-route semantic score, adjudicate every required evidence item on the routes used for that estimate; a convenience sample of failing excerpts is insufficient. Freeze the rubric on development examples and evaluate it on held-out items. Because the existing traces have already been inspected, this is a declared post-hoc audit, not a preregistered re-evaluation of the original primary endpoint.

This paper reports no semantic pass rate: the adjudication has not been performed. The practical value of the proposal is to make the next measurement answer a precise question: how often does strict trace failure indicate unsupported reasoning, and how often does it indicate a representation defect? The distinction matters for both reliability claims and engineering effort.

3.3 Dependence, uncertainty, and error propagation

The benchmark contains 300 underlying projects, not 5,094 independent enterprises. Multiple gates within a project share facts and handoffs; multiple models review the same project. The released case-cluster and paired analyses preserve those dependencies for the reported intervals. Those intervals still condition on the generated population and one retained trajectory per model and case. They do not measure uncertainty over organizations, model versions, or repeated inference runs. Temperature zero does not supply a replication study.

Sequential handoffs allow an upstream output to affect a later review. They make propagation observable in traces, but this single handoff condition does not identify a causal propagation effect. A controlled comparison would run the same cases with agent handoffs, reference handoffs, and no prior-review content, with matched resource budgets and repeated trajectories. Report downstream changes conditional on the upstream error class, rather than treating every repeated mention of one defect as an independent safety event.

The original gate definition is a useful strict endpoint; it is not the only endpoint an operating organization would choose. A deployment evaluation should separately register false approvals, missed critical findings, evidence validity, unauthorized actions, cycle time, and human work. Acceptance thresholds should follow the consequence of each error. The present study supplies observations for several of those outcomes, but no enterprise has validated its contract as a sufficient production acceptance test.

4 The role of forward deployed engineers

Here, a forward deployed engineer (FDE) denotes an engineer working with an organization’s domain owners to integrate and operate a review system. The contribution of the role is not assumed from its title. It consists of deliverables that make a review contract executable, observable, and maintainable. The benchmark already supplies many of these prerequisites; in an enterprise they must be built and kept correct.

4.1 From a generated contract to an operational service

The first task is policy formalization. A specialist and the policy owner must agree which decision conditions are mandatory, which permit an exception, and which require interpretation. The implementation should record version, scope, effective date, and responsible authority for each rule. A model’s explanation is not a substitute for resolving contradictory policies. Ambiguities discovered during formalization are findings about the organization’s process and should remain visible in the deployment record.

The second task is evidence integration. Synthetic snapshots make facts convenient to read; real evidence comes from versioned repositories, identity services, ticketing systems, monitoring, contracts, and people. Each connector needs access control, freshness checks, stable source identifiers, and recorded observations. If two sources disagree, the system needs an escalation or precedence rule rather than silently selecting whichever supports approval. The cost of producing a reliable snapshot belongs in the deployment, even if the subsequent model review becomes almost trivial.

The third task is authority. Recommendation, conditional approval, and execution are distinct capabilities. The reviewer must act within a versioned mandate whose scope includes the project, action, and any conditions. An exception mechanism must retain the unresolved finding and identify who accepts it. Approval status must not be inferred solely from a fluent rationale. The benchmark’s rejected conditional-approval attempts make this distinction concrete: a proposed call can be invalid even when a later submitted decision is correct.

The fourth task is a shared case state. Later gates need to know which obligations remain open, which evidence is stale, and which prior opinions have been superseded. Passing an unstructured summary alone can lose conditions; copying every historical record can create contradictions. An implementation should retain typed decisions, evidence references, policy versions, unresolved obligations, and the causal relationship between reviews. Gates can be introduced incrementally while using the same state and interface design.

Table 9: Proposed FDE deliverables and independently checkable acceptance evidence. These are implementation requirements, not deployment outcomes measured in this benchmark.

Deliverable	Concrete artifact	Acceptance evidence
Policy contract	Versioned rules, exception scope, output schema	Specialist adjudication of held-out cases and documented disagreements.
Evidence interface	Connectors, timestamps, source IDs, read log	Tests of stale, missing, contradictory, and access-restricted records.
Authority boundary	Mandate registry and enforced action checks	Invalid calls rejected; accepted actions trace to an active mandate.
Connected case state	Findings, obligations, decisions, dependency graph	Replayed handoffs preserve open issues and invalidate stale approvals.
Review service	Agent, validator, escalation and recovery behavior	Held-out quality results including resource exhaustion and provider failure.
Operating account	Human activity ledger and support allocation	Measured effort includes specialists, engineers, suppliers, and rework.

4.2 Failure-informed engineering priorities

The observed failure classes suggest testable interventions. An evidence serializer could attach verbatim spans and stable source identifiers directly from the read log, avoiding a model’s reconstruction of exact quotations. A consistency check could flag a submitted approval whose rationale explicitly asks for rework. A finalization contract could require abstention or escalation when resource limits prevent a supported answer, instead of permitting a placeholder approval. A handoff validator could ensure that an unresolved upstream prohibition cannot silently disappear from the final review.

These interventions could reduce particular failures, but their effects are not measured here. Each should be evaluated as a new system version on held-out cases, recording any new failure modes and human burden. A validator that catches an error but routes every case to a specialist may improve decision safety without reducing labor. A serializer that makes quotes exact may raise strict CSR without improving substantive judgment. The outcome attributed to an intervention should match what it changes.

Operational performance is a property of the complete system: policy owners, evidence systems, model, validators, authorization controls, and exception handling. The FDE’s recurring work is part of that system. Initial integration hours are a transition investment; repeated policy changes, connector repairs, evaluations, incident investigations, and model migrations are recurring work. Moving them into an external supplier does not remove them from the DGF labor boundary. Separate reporting of build effort and recurring effort enables both an annual capacity estimate and a cumulative payback analysis.

4.3 Testing the implementation hypothesis

We hypothesize that a connected implementation with shared evidence and obligations can avoid some re-reading, re-entry, and reconciliation across review teams. The relevant comparison is with locally optimized gate tools at comparable project mix, authority, quality, and automation scope. Deployment can remain incremental in both conditions. Measure duplicated evidence collection, downstream correction, specialist review hours, recurring support, and total quality-constrained human work. Assign implementation approaches across comparable workflows where feasible, or use a staged comparison with explicit confounding controls.

An improvement would support the value of the shared implementation, not prove that a particular job title caused it. No FDE deployment is included in the current experiment. Likewise, the idea that governance will automate before other business functions is a comparative hypothesis: it requires matched functions, comparable outcome definitions, and observed deployment times. The present results do not establish a universal ordering. They identify a concrete candidate for further study because much of the review contract can be specified and inspected.

5 From task success to required human work

5.1 A minimal accounting identity

Automating a review can remove reading, comparison, documentation, and coordination work. It can also add exception handling, monitoring, correction, and maintenance. To determine the net effect, first define the same organizational boundary and project volume in both regimes. Let V be reviews per measurement period and h_0 the baseline mean human hours per review. Let p be the fraction routed to substantive human exception handling, h_E the total human hours on each such review, h_S the human hours on a standard-path review, and h_W additional rework averaged over all reviews. Let B_0 and B_1 denote case-independent support hours before and after deployment. Then

$$L_0 = Vh_0 + B_0, \tag{1}$$

$$L_1 = V\{ph_E + (1 - p)h_S + h_W\} + B_1, \quad N_j = L_j/K \quad (j = 0, 1). \tag{2}$$

Here K is a fixed useful-hours convention per FTE per period. The same convention is used in both regimes. Exception time includes any preliminary review of those same cases. Rework excludes activities already counted in exception time, ordinary review, or separately counted repeat visits. Support includes internal and external work within the registered function.

Equation (2) is accounting, not an estimated response to AI adoption. With consistent inputs it is equivalent to adding the underlying activity hours directly. We make no hypothesis that it outperforms an algebraically identical comparator. Its purpose is to expose omissions and make an

empirical claim about savings auditable. Gate success, approval rate, and the share of tool calls completed are not substitutes for any of these human-time variables.

5.2 Why remaining cases can retain disproportionate work

Let w denote the exception group’s share of baseline case-handling hours, under a fixed partition of the same case population. Let a be its post-deployment effort divided by its own baseline effort, and u the corresponding ratio for the standard group. Put $r = h_W/h_0$ and $b_j = B_j/(Vh_0)$. For positive baseline group means,

$$R := \frac{N_1}{N_0} = \frac{wa + (1 - w)u + r + b_1}{1 + b_0}. \quad (3)$$

This is the same labor decomposition as the companion manuscript’s $\phi\eta + (1 - \phi)\mu + r + b_A$ numerator, with a smaller, non-reused notation set. All quantities are defined once in Appendix A. If a group’s baseline mean is zero, use the hours identity instead of an undefined effort ratio.

The exception case share p and its baseline labor share w need not coincide. A small group of difficult projects may retain much of the original work. The relation follows from the covariance of routing and baseline effort; Appendix B gives the short proof. This is a property of selection, not evidence that the benchmark’s failing cases are the costly enterprise cases. Human timing data are needed to establish that relationship.

Table 10: One transparent sensitivity illustration, not fitted to DGF-Bench. Both rows set $w = 0.40$, $b_0 = 0$ and use the same case partition.

Assumed regime	a	u	r	b_1	R	FTE from 100
Low residual effort	0.25	0.05	0.02	0.05	0.20	20
High residual effort	1.50	0.50	0.10	0.10	1.10	110

The difference in the table is produced by the stated assumptions. It is not a discovered effect, a fitted forecast, or an estimate obtained from the model success rates. Its use is to show exactly what would need to be measured: assistance on exceptions, residual standard review, additional rework, and recurring support. The long manuscript’s multiple workforce scenarios are omitted here because repeating such arithmetic does not add empirical evidence.

5.3 Implications for staffing and employment

If validated review automation lowers L_1 at comparable output and quality, fewer workload-equivalent FTE are required for those activities. This is a mechanism through which work previously performed by employees can be replaced. A company may then reduce staffing, avoid new hires, redeploy people, or increase reviewed volume. Those are observable choices, not identical outcomes. Required hours, allocated staffing, headcount, and hiring should be reported separately so that the workforce claim can be evaluated rather than inferred from a technology demonstration.

An organization could reduce headcount while leaving unmet review demand or transferring unpaid work to another team. Conversely, it could preserve headcount while governing more projects with the same effort. The primary comparison therefore standardizes volume and case mix, retains quality and service requirements, and includes shared supplier support. Initial implementation effort should also be reported for the investment decision, even when the primary endpoint concerns recurring annual work.

The benchmark supports the technical premise that models can execute many represented tasks. It contains no human hours and cannot estimate R . The author’s projection of an 80% reduction in required DGF FTE by 2033 is a hypothesis about a possible deployment outcome, rather than the numerical complement of any model’s accuracy. The next section makes its measurement conditions and calendar explicit.

6 A feasible validation program

The empirical gaps require additional observations. The following studies are proposed; no human participants, enterprise cohort, or independent semantic adjudication have been collected for this paper. Their protocols should be finalized, independently reviewed, and registered before new outcome collection. Existing benchmark inspections and design choices must be disclosed as prior information. The companion files in `paper2/protocols/` provide an operational starting point rather than a claim of preregistration.

6.1 Human and deterministic baselines

A fair baseline must first specify which ability is being compared. A human with only Word documents and a model with authoritative structured facts are not solving the same information problem. The first comparison should therefore give qualified reviewers and agents the same policy, accessible evidence, mandate information, output requirements, and case population. Humans can use a form that captures structured outputs; record form-entry time separately from investigation and deliberation. The exact-quotation requirement should be explained to both groups, while a separate semantic evaluation tests support independently of serialization.

Evaluate at least three conditions: unaided human review, agent review, and human review of an agent-produced draft. Add a deterministic policy engine that reads only the permitted facts and mandates, without access to evaluator-only answers. This comparator tests whether the explicit rule system itself accounts for disposition performance. Keep extraction from unstructured documents as a separate experimental condition. An oracle that reads the answer key can check the harness, but cannot serve as a substantive baseline.

Recruit domain-qualified participants, record relevant experience, and train them on development cases excluded from evaluation. Assign held-out cases with counterbalanced order so that a participant does not review the same case once unaided and again with an agent draft. Preserve route-level context when measuring complete routes. Independently adjudicate answer quality without revealing the review condition where practical. Report compensation, missing sessions, exclusions, and any identifiable information retained under the study’s ethics requirements.

Primary outcomes should include independently adjudicated disposition quality, false approval and critical-miss rates, semantic evidence support, and active human minutes. Total cycle time and exact-format conformity are secondary outcomes. For the assisted arm, count time verifying the draft, finding errors, rewriting outputs, and escalating issues; a short initial reading time alone is not an efficiency gain. Record time spent by other specialists who receive questions from the assigned reviewer.

The number of cases and reviewers should follow a prospective power and precision analysis using a stated noninferiority margin, expected paired disagreements, and participant/case clustering. Three hundred synthetic cases do not by themselves determine the sample size needed to establish rare-error safety or workforce effects. A small pilot may estimate variance and operational burden, but its outcomes must not be used repeatedly to choose a favorable primary metric. A confirmatory set should remain untouched until the analysis is frozen.

6.2 Independent cases, scaffold removal, and repeated trials

The generator and evaluator share an underlying rule system. An independent set should be written or adjudicated by specialists who did not construct those rules, using policies and documents whose ambiguities are documented. Specialists should be allowed to conclude that a case is underdetermined. Such cases test escalation rather than forcing a single convenient answer. Keep both policy disagreements and unambiguous implementation errors visible.

On a held-out synthetic set, compare full structured facts with a document-only condition, then

introduce stale evidence, conflicting versions, and incomplete records using frozen transformations. Validate that the necessary facts remain recoverable when the task is intended to be solvable. A drop after hiding REVIEW_FACTS would quantify dependence on the scaffold; it would not by itself distinguish extraction difficulty from reasoning difficulty. Add an extraction-only measurement so that these components can be examined separately.

Repeat trajectories for every selected model–case pair using a predeclared repeat count, reporting all attempts rather than selecting the best. Record provider, timestamps, resolved model identifiers, inference settings, resource limits, and retries. Separate sampling over projects from variability over repeated executions. If costs constrain replication, use a stratified case sample whose selection is fixed before outcomes and retain its inclusion probabilities. Expand the endpoint set to test whether the observed ranking survives a broader model selection; the current three endpoints are not a population sample of AI systems.

Adversarial document content and prompt injection should form a separately declared condition. Such tests concern whether the reviewer maintains its rules and authority while reading untrusted evidence, as distinct from applying a trusted snapshot correctly. A favorable score on the current dataset provides no estimate of that failure rate.

6.3 Enterprise deployment: quality and labor together

Field evaluation should begin with shadow review: the system produces outputs while existing authority remains unchanged. Independently compare the outputs and measure the human work needed to correct them. Shadow operation establishes review performance and integration burden; it does not by itself demonstrate savings because it may add a parallel review. A later controlled stage must measure actual substitution of defined activities under the approved operating design.

Where feasible, randomize comparable workflows or teams to staged adoption and retain a contemporaneous comparison group. If randomization is not feasible, register a design using pre-adoption trends, case-mix adjustment, and explicit confounders such as policy changes, reorganization, demand changes, and other automation. A before–after reduction alone establishes a temporal change, not attribution to the agent. Report both the volume-standardized effect and the actual workload trajectory.

The activity ledger should include evidence preparation, ordinary review, exception handling, remediation coordination, quality checks, incident work, and recurring system support. Shared platform teams and supplier operators must be counted once, then allocated under a frozen rule. Unobserved external effort is missing, not zero. Maintain an obligation register so that apparent savings cannot be achieved by leaving corrective work unfinished or pushing it beyond the measurement window. The observation period must include scheduled maintenance and policy updates.

Quality requirements should be registered with domain owners before deployment: admissible decision error, critical misses, unauthorized actions, evidence validity, and service delay. Noninferiority on an average score is insufficient if it hides unacceptable critical failures. The primary labor reduction is interpretable only while those requirements are met. If a site reverts to manual review, its recovery and manual effort remain part of the outcome.

6.4 Repairing the 2026–2033 calendar

The original dated hypothesis is that required DGF FTE will be at most 20% of the 2026 baseline by 20 September 2033, at comparable governed output, quality, and service, including recurring support. Its baseline is the twelve-month period ending 20 September 2026; its endpoint is the twelve-month period ending 20 September 2033. As of this manuscript, the baseline window has already closed and no cohort is enrolled. It cannot be opened prospectively after the fact.

There are two legitimate paths. The first is a historical-baseline study: enroll organizations with auditable records covering the original window, reconstruct the activity boundary, validate time attribution, and freeze the reconstruction before examining endpoint outcomes. Register record eligibility and missingness rules before selecting sites. Include an audit of contracts, timesheets, case volumes, supplier effort, and support allocation. Retrospective estimates and their uncertainty must be disclosed; missing history cannot be replaced by recollection without changing the evidential status of the comparison.

The second is a prospective cohort with a new baseline. Fix a registration date t_0 , measure twelve months of baseline activity, and specify a follow-up period before observing its outcomes. It can test subsequent labor reduction, but it is a new dated hypothesis. A seven-year comparison from a later baseline has a later endpoint. A comparison ending in 2033 instead has a shorter interval. Neither should be reported as confirmation of the original 2026–2033 prediction.

Table 11: Calendar rules. Neither cohort exists at publication; the table specifies eligible designs.

Design	Required baseline	Interpretation
Original 2033 test	Auditable reconstruction of the year ending 20 September 2026	Tests the dated ratio for the covered population if endpoint and quality data are complete.
New prospective cohort	Twelve months collected after registered t_0	Tests a newly stated baseline and horizon; cannot silently replace the original prediction.
Insufficient records	Missing historical hours or unmeasured material supplier effort	Original dated test remains untestable for that population; no confirmation claim.

For the original test, estimate a volume- and case-mix-standardized final/baseline labor ratio including support. Register organizational weighting, clustering, attrition treatment, and uncertainty from time reconstruction. Confirmation requires the upper bound of the registered uncertainty interval to be at most 0.20 and all quality and service criteria to pass. A lower bound above 0.20 contradicts the threshold for that population. An interval spanning the threshold is inconclusive; an estimate below it with missing material labor is not confirmation. Failure to enroll an eligible cohort leaves the forecast untested rather than moving its date.

A convenience cohort supports claims about its participating organizations. Ecosystem-wide inference requires a defensible sampling frame, coverage of functions and suppliers, and appropriate weighting. Employment consequences also require reporting staffing and headcount separately. The causal claim that AI produced the change requires the comparison design in addition to observing the threshold. These distinctions specify what data must be collected; they do not weaken the possibility of a large reduction when those data support it.

7 Conclusion

DGF-Bench shows that the tested models can complete many specified governance reviews under explicit rules and accessible structured facts. Gemini attains 94.98% strict gate success and 230 completely conforming routes among 299 evaluable projects. Its dispositions are correct throughout that evaluable sample; its strict failures concern evidence conformity. Luna and DeepSeek also show substantial task capability, with lower strict success and observed substantive errors. The released traces permit these outcomes to be inspected instead of reducing them to a single demonstration or a single accuracy number.

These results support the technical prospect of automating a substantial share of the review tasks represented here. If comparable performance can be achieved on operational evidence with valid authority and acceptable assurance costs, automation could replace work currently performed

by employees and reduce the staffing required for DGF activities. That is the task-substitution hypothesis advanced in this paper. An FDE contributes by making the policy, evidence, mandates, handoffs, recovery, and human-work measurement function as one operating system. The recurring work required to maintain it belongs in the result.

The next decisive evidence is a comparison with qualified humans and a deterministic rules baseline, independently adjudicated evidence support, scaffold-removal tests, repeated trajectories, and measured human hours in deployment. The proposed studies are designed to separate a formatting repair from better judgment, task automation from merely parallel review, and a staffing cut from lower required work. The 80% FTE reduction by 2033 remains an explicit projection whose original test requires credible 2026 records. Complete governance automation is not established as a law. The substantive claim is ambitious and testable: reliable execution of review contracts may substantially reduce the human work required to govern projects.

Data, code, and research status

The [DGF-Bench repository](#) hosts the public materials. The release [dgm-bench-300-20260923](#) contains the full recorded run (17,700 original files), prepared dataset (27,947 files), and benchmark source snapshot (1,099 files). It includes Word documents, architecture diagrams, simulated records, policies, traces, scores, usage ledgers, and historical failed attempts. Checksums and per-file inventories preserve provenance. Evaluator-only references are released for audit; they were not exposed by agent tools during the run.

The final data tables and reproduction instructions are in [research/2026-09-dgm-bench/](#). Offline reproduction uses the frozen source archive, not an assumption that the evolving default branch is identical. The earlier 15-project pilot is separately available, including its Word documents, under [experiments/run_20260922_180730_063947/](#); its outcomes are not pooled into this study. The second manuscript sources are in [paper2/](#). Its inherited benchmark tables and figures retain the original measurements. No paid inference, human study, semantic adjudication, or enterprise enrollment was performed to create this shorter manuscript.

The author is Jeremy Canale, contact@jeremycanale.com, <https://www.jeremycanale.com>, <https://www.linkedin.com/in/jcanale13>. The workflow specification is author-designed; no confidential enterprise documents or employment outcomes are represented as observations. AI assistance was used to prepare and revise this manuscript and its supporting software; this is not independent expert adjudication. The author remains responsible for checking the claims, artifacts, and interpretation before submission.

References

- Daron Acemoglu and Pascual Restrepo. Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30, 2019. <https://doi.org/10.1257/jep.33.2.3>.
- Lisanne Bainbridge. Ironies of automation. *Automatica*, 19(6):775–779, 1983. [https://doi.org/10.1016/0005-1098\(83\)90046-8](https://doi.org/10.1016/0005-1098(83)90046-8).
- Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *The Quarterly Journal of Economics*, 140(2):889–942, 2025. <https://doi.org/10.1093/qje/qjae044>.
- Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. arXiv:2406.13352v3, 2024. <https://arxiv.org/abs/2406.13352v3>.
- Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H. Laradji, Manuel Del Verme, Tom Marty, David Vazquez, Nicolas Chapados, and Alexandre Lacoste. WorkArena: How capable are

web agents at solving common knowledge work tasks? In *Proceedings of the 41st International Conference on Machine Learning*, PMLR 235:11642–11662, 2024. <https://proceedings.mlr.press/v235/drouin24a.html>.

A Notation and accounting boundary

Table 12: Workload notation. Each symbol has a single role within this manuscript.

Symbol	Definition
V	Reviews per period, held comparable between regimes.
h_0	Mean baseline case-handling hours.
p	Share of cases assigned to human exception handling.
h_E, h_S	Post-deployment human hours per exception and standard case.
h_W	Additional rework hours per review, averaged over all cases.
B_0, B_1	Case-independent support hours in baseline and deployment regimes.
L_0, L_1	Total human hours per period within the same boundary.
K	Fixed useful-hours convention per FTE per period.
N_0, N_1	Required workload-equivalent FTE; $N_j = L_j/K$.
w	Exception group’s share of baseline case-handling labor.
a, u	Post-deployment effort ratios against each group’s own baseline.
r, b_0, b_1	Rework and support normalized by baseline case-handling effort.
R	Required FTE ratio at comparable volume and quality.

The dimensional identity is valid at $p = 0$ or $p = 1$ using only the group that exists. Group-relative ratios require positive baseline group means. The constraint $0 \leq w \leq 1$ does not constrain a or u to be at most one: review or exception work can increase after deployment. Fixed support is not eliminated by multiplying every term by a volume adjustment. If volumes change, separate fixed support from variable review effort before standardization.

An activity should be counted exactly once regardless of departmental ownership or supplier boundary. For example, a specialist’s correction after an agent approval is either exception handling or additional rework under the declared coding rule, not both. A shared evaluation team’s hours are counted once before allocation across workflows. A repeat gate visit may be included in review volume or in per-project effort; the study must choose a consistent basis. Elapsed waiting time and active human time are distinct, although exclusive monitoring may consume human capacity. Record the rule used for that case.

B Residual selection: the short result

Proposition 1. *Let $T \geq 0$ be integrable baseline handling time with mean $h_0 > 0$. Let J indicate assignment to human exception handling, $p = \Pr(J = 1) > 0$, and $e(T) = \Pr(J = 1 \mid T)$. The exception group’s baseline labor share satisfies*

$$w = \frac{\mathbb{E}[TJ]}{h_0} = p + \frac{\text{Cov}(T, e(T))}{h_0}.$$

Consequently it exceeds the case share exactly when routing is positively associated with baseline effort. If e is nondecreasing, this covariance is nonnegative.

Proof. Conditional expectation gives $\mathbb{E}[TJ] = \mathbb{E}[Te(T)]$ and $\mathbb{E}[e(T)] = p$, giving the identity. For independent copies T, T' , $2 \text{Cov}(T, e(T)) = \mathbb{E}[(T - T')(e(T) - e(T'))]$. The integrand is nonnegative when e is nondecreasing. $\square$

No claim about the conditional variance of exception times follows from the proposition. Estimating w requires baseline hours under a frozen routing partition, not just a count of cases later failed

by a model. Changes in the routing rule require a new partition or a documented reweighting. The identity supports measurement of concentration of remaining work; it does not predict its size without those measurements.

C Verification and follow-up record

The original strict outcomes remain the primary results of the existing experiment. A future semantic audit should retain item identifiers, observed evidence paths, cited claims, independent labels, disagreement reasons, adjudicated labels, and sampling probabilities. Release the rubric and analysis version with the results. Keep the lexical and semantic scores side by side so that differences can be traced to particular findings rather than only to a revised headline rate.

The field protocol must record registration identifier, enrollment date, eligible population, baseline provenance, operating boundary, quality thresholds, primary labor outcome, uncertainty procedure, and missing-data rules before outcome analysis. None of those identifiers is invented for this manuscript. If historical records cannot support the 2026 baseline, a new prospective study must state its own dates and hypothesis explicitly.

The repository's `paper2/REVIEW_RESPONSE.md` maps the methodological criticisms to editorial changes and outstanding empirical work. The `paper2/verify_manuscript.py` utility checks the inherited tables, figure hashes, headline result counts, and the declared sensitivity illustration against the released sources. These checks concern internal consistency and provenance, not independent validation of the underlying enterprise rules.